PROPER BASE CHANGE FOR THE DERIVED PUSHFORWARD OF A COMPLEX ON TOPOLOGICAL SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. This card was re-admitted after a gate audit: it had been excluded only because the chapter's retained preamble previously lacked the reference/slogan/history environments. The shared preamble now carries them (appended 2026-10-12), so the card uses that file as its preamble with no per-card copy.

Editorial note. The author's own bibliographic remark, which sits in a reference environment at the head of the theorem and points at the original exposition, is preserved in the source span and is not typeset because the project's own preamble defines reference environments as comments; no bibliography excerpt is needed.

Editorial note. The label theorem-proper-base-change is defined only in cohomology.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Fibre formula for stalks of the derived pushforward under closedness, separatedness and quasi-compact fibres. Same chapter, Lemma lemma-proper-base-change, invoked at proof lines 3128-3135. Supplies the identification of both sides with derived global sections over the fibres.
- Characterization of properness and stability of the relevant hypotheses under base change. Topology, Theorem topology-theorem-characterize-proper and Lemma topology-lemma-base-change-separated, invoked at proof lines 3124-3127. Supplies that the base changed map is proper with quasi-compact fibres and that both maps are closed.
- The base change map for derived pushforwards. Same chapter, Lemma lemma-base-change-map-flat-case, named in the statement at line 3117. Supplies the map whose bijectivity is the content of the theorem.

Source reference. Original lemma 02N7: sources/stacks-cohomology/pinned-original-source.tex, source lines 2977–2977. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 09V5: sources/stacks-cohomology/pinned-original-source.tex, source lines 3059–3059. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0CY4: sources/stacks-chow/tags.txt, source lines 16229–16229. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original theorem 005R: sources/stacks-chow/tags.txt, source lines 219–219. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Theorem 0.1 (Proper base change)**.** *Consider a cartesian square of topological spaces*

$$\begin{array}{ccc} X' = Y' \times_Y X & \xrightarrow{g'} & X \\ \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y \end{array}$$

Assume that f is proper. Let E be an object of $D^+(X)$. Then the base change map

$$g^{-1}Rf_*E \longrightarrow Rf'_*(g')^{-1}E$$

of Lemma 02N7 is an isomorphism in $D^+(Y')$.

Proof. Let $y' \in Y'$ be a point with image $y \in Y$. It suffices to show that the base change map induces an isomorphism on stalks at y'. As f is proper it follows that f' is proper, the fibres of f and f' are quasi-compact and f and f' are closed, see Topology, Theorem 005R and Lemma 0CY4. Thus we can apply Lemma 09V5 twice to see that

$$(Rf'_*(g')^{-1}E)_{y'} = R\Gamma((f')^{-1}(y'), (g')^{-1}E|_{(f')^{-1}(y')})$$

and

$$(Rf_*E)_y = R\Gamma(f^{-1}(y), E|_{f^{-1}(y)})$$

The induced map of fibres $(f')^{-1}(y') \to f^{-1}(y)$ is a homeomorphism of topological spaces and the pull back of $E|_{f^{-1}(y)}$ is $(g')^{-1}E|_{(f')^{-1}(y')}$. The desired result follows. □